# LAURA RODRIGO HERNANDO

Full-Stack Developer · Angular · Python/FastAPI · AI Engineer

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## PROFILE

Full-stack developer with 3+ years of experience building data visualisation web applications with Angular, Python/FastAPI and PostgreSQL/PostGIS: interactive maps, charts, REST APIs and Docker and CI/CD deployments. I have worked on more than 8 applications, from the user interface to the database, and built an internal application end to end with Keycloak authentication. I bring generative AI into my work through data extraction with local vision models, RAG systems and coding agents.

## TECHNICAL SKILLS

**Frontend:** Angular, TypeScript, HTML/CSS, ng-zorro, PrimeNG, Clarity, Leaflet, ECharts, Plotly.js, GSAP, That Open (IFC/BIM), i18n, Figma

**Backend & data:** Python, FastAPI, .NET, PostgreSQL + PostGIS (GeoJSON), Celery, data migrations (CSV/JSON), DBeaver, uv, Poetry

**Security:** Keycloak (setup, administration and 2FA), JWT, cross-application roles and permissions

**DevOps & quality:** Git, CI/CD, Docker, development and production environments, SonarQube, Cypress, pytest, Linux

**Applied AI:** Local LLMs (Ollama), vision models, RAG (LlamaIndex, LangGraph), LLM API integration (OpenAI, Anthropic), coding agents (Claude Code, Codex)

**AI foundations:** Machine learning and deep learning, neural networks, transformers and attention, embeddings; PyTorch, pandas, NumPy, Hugging Face, LangChain, Langfuse

**Ways of working:** Scrum, multidisciplinary teams, mentoring interns and juniors

## PROFESSIONAL EXPERIENCE

**Full-Stack Developer** · CARTIF, Boecillo (Valladolid), Spain *Jun 2023 – present*

- Develop 8+ data visualisation and analysis web applications with Angular and Python/FastAPI, from the user interface to the database.
- Turn prediction models into web tools: parameter configuration, asynchronous execution (Celery) and results on maps (Leaflet) and charts (ECharts, Plotly.js).
- Design REST APIs and geospatial databases (PostgreSQL/PostGIS) and contribute to architecture decisions and frontend-backend contracts.
- Ensure quality and continuous delivery with E2E (Cypress) and backend (pytest) tests, SonarQube analysis, Docker containers and CI/CD pipelines.
- Work in Scrum within multidisciplinary teams, and train and mentor interns and junior developers.

### SELECTED PROJECTS

**Internal tools catalogue (in progress)** · Full-stack · technical owner

Designed and built the whole application: PostgreSQL/PostGIS data model, FastAPI REST API and Angular frontend. Set up Keycloak with two-factor authentication and embedded, with shared permissions, another CARTIF application I built with a colleague and adapted for the integration.

**Receipt and invoice extraction with AI** · Backend & AI · team of 2

FastAPI API that classifies photos of receipts and invoices with a local vision model (Ollama) and extracts category, amount and currency as structured JSON, rejecting unreadable or invalid documents. Data never leaves the company's infrastructure.

**Climres (in progress)** · Backend

Geospatial PostgreSQL/PostGIS database for GeoJSON data, FastAPI REST API and Python migration scripts to integrate CSV and JSON data.

**Smarteestory (in progress)** · Frontend

Building digital twin: IFC/BIM model viewer (That Open), sensor data from an external API and a full document manager.

**Spandam and RethinkAction** · Frontend

Decision-support tools with interactive maps, charts and asynchronous calculations. On Spandam I proposed design improvements on top of Figma and built the bilingual interface and the Cypress E2E tests.

**Other frontend projects** · Hydro4u and Rehouse (in progress), Build Space, Nevermore

Interfaces with Angular, Leaflet, ECharts and component libraries (ng-zorro, PrimeNG, Clarity): prediction model visualisation, document management, collaborative projects and mapping.

**Web Developer Intern (training contract)** · CARTIF, Boecillo (Valladolid), Spain *Mar 2023 – May 2023*

- Built end-to-end features for the corporate intranet with Angular, .NET and PostgreSQL, working with secure JWT-based authorisation and CI/CD deployment.

## PERSONAL PROJECTS

**Interactive portfolio with a RAG assistant**

Personal website with a 3D galaxy (Next.js, React Three Fiber) and an assistant that answers questions about my profile: RAG with LangGraph and hybrid search, compatible with local (Ollama) and cloud models (OpenAI, Anthropic).

**Educational RAG chatbot (in progress)**

Assistant for an educational project built on a custom LlamaIndex RAG system, designed to run on both local and cloud models.

## EDUCATION & TRAINING

**Fundamentals and tools for generative AI (in-company training)** · Universidad Politécnica de Madrid *2026*

Transformers and attention, model training (overfitting, underfitting, data augmentation), Hugging Face, LangChain, LangGraph, Langfuse and RAG.

**Machine Learning and Deep Learning (in-company training)** · Universidad de Valladolid *2025*

Mathematical foundations and algorithms of ML and DL: neural networks, backpropagation, activation functions, embeddings and transformers. Hands-on work with PyTorch, pandas and NumPy.

**Learn FastAPI from scratch** · Udemy *2025*

**Higher Vocational Diploma in Web Application Development** · IES Galileo, Valladolid *2021 – 2023*

## LANGUAGES

Spanish: native · English: professional working proficiency

## AVAILABILITY

Remote, asynchronous work, either employed or as a freelancer on a fixed commitment (a set number of hours per week for an agreed period).